Conley-Zehnder approach to Arnold conjecture

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Abstract

This is a note for my lectures on additional topics given in the exercise session for the course **Floer Homology** SS2024 at the Universität Augsburg.

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1 Hamiltonian system and action functional

1.1 Hamiltonian system in $\mathbb{R}^{2n}$

Let $H(t,x) \in C^\infty(S^1 \times M)$ be a time-dependent Hamiltonian function on $(M,\omega) = (\mathbb{R}^{2n}, \omega_0)$ with $\omega_0 = d\lambda_0$ and

$$\lambda_0 = pdq = \sum_{i=1}^{n} p_i dq_i$$

in terms of coordinate

$$(q,p) = (q_1, \ldots, q_n, p_1, \ldots, p_n) \in \mathbb{R}^{2n}.$$

The corresponding Hamiltonian vector field X_H is determined by

$$-dH = \omega_0(X_H, \cdot).$$

The standard ω_0-compatible almost complex structure

$$J_0 = \begin{pmatrix} 0 & -I_n \\ I_n & 0 \end{pmatrix}$$

yields

$$\langle \cdot, \cdot \rangle = \omega_0(\cdot, J_0 \cdot), \qquad \omega_0(\cdot, \cdot) = \langle J_0 \cdot, \cdot \rangle.$$

The Hamiltonian vector field can be rewritten as symplectic gradient field

$$X_H = J_0 \nabla H$$

where ∇ is the gradient with respect to Euclidean metric $\langle \cdot, \cdot \rangle$. The Hamiltonian system is the non-linear ODE:

$$\dot{x} = J_0 \nabla H(t, x(t)). \tag{1}$$

For the non-linear Hamiltonian system (1) in $\mathbb{R}^{2n}$, its linearization at a solution $x(t)$ of (1) reads

$$\dot{\xi}(t) = J_0 \nabla^2 H(t, x(t)) \xi(t), \qquad \xi \in C^\infty(S^1, \mathbb{R}^{2n}). \tag{2}$$

We now describes several important types of Hamiltonian systems:

- Suppose $B(t)$ is a loop of symmetric matrices,

$$H(t, x(t)) = \frac{1}{2}\langle B(t)x, x\rangle$$

is called **periodic quadratic Hamiltonian**. Such type is refereed to as **linear Hamiltonian system**.

- $H \in C^2(S^1 \times \mathbb{R}^{2n})$ is **asymptotic linear** if there exists a loop of symmetric matrices $B_\infty(t)$ s.t.

$$\lim_{\|x\|\to\infty} \frac{\|\nabla H(t,x) - B_\infty(t)x\|}{\|x\|} = 0 \tag{3}$$

uniformly with respect to t. The linear system $\dot{\xi}(t) = J_0 B_\infty(t)\xi(t)$ is called the **linearized system at infinity**.

1.2 Action functional and L^2-gradient

The Hamiltonian action functional $\mathcal{A}_H : C^\infty(S^1, \mathbb{R}^{2n}) \to \mathbb{R}$ is defined by

$$\begin{aligned}\mathcal{A}_H(x) &= \int_{S^1} x^*\lambda_0 - \int_{S^1} H(t, x(t))dt \\ &= \int_{S^1} \left(-\frac{1}{2}\langle J_0\dot{x}(t), x(t)\rangle - H(t, x(t))\right) dt \\ &= \int_{S^1} (\langle p(t), \dot{q}(t)\rangle - H(t, p(t), q(t)))dt\end{aligned}$$

where $x(t) = (p(t), q(t))$. The differential of $\mathcal{A}_H$ at x is

$$d\mathcal{A}_H(x)(\xi) = \int_{S^1} \langle J_0\dot{x}(t) + \nabla H(t, x(t)), \xi(t)\rangle dt \qquad \forall \xi \in C^\infty(S^1, \mathbb{R}^{2n}).$$

The critical points of $\mathcal{A}_H$ are solutions of Hamiltonian system (1).

Under the identification $(\mathbb{R}^{2n}, J_0) \cong (\mathbb{C}^n, i)$, the Fourier series expansion for a smooth loop $x : S^1 \to \mathbb{R}^{2n}$ reads

$$x(t) = \sum_{k\in\mathbb{Z}} e^{2\pi k t J_0}\hat{x}_k, \qquad \hat{x}_k \in \mathbb{R}^{2n}.$$

The action functional takes the form

$$\mathcal{A}_H(x) = a(x) - b(x) \tag{4}$$

where

$$b(x) = \int_{S^1} H(t, x(x))dt \tag{5}$$

and the leading term of $\mathcal{A}_H$ is a quadratic form

$$a(x) = \frac{1}{2}\langle Ax, x\rangle_{L^2} = \pi \sum_{k\in\mathbb{Z}} k\|\hat{x}_k\|^2 \tag{6}$$

with $A = -J_0\frac{d}{dt}$ regarded as an unbounded self-adjoint operator on $L^2(S^1, \mathbb{R}^{2n})$ whose domain is $\mathcal{D}(A) = H^1(S^1, \mathbb{R}^{2n})$. It has closed image, a pure point spectrum and a compact resolvent. The spectrum reads

$$\begin{aligned}
\sigma(A) &= \sigma_{\text{disc}}(A) \sqcup \sigma_{\text{ess}}(A) \\
\sigma_{\text{disc}}(A) &= 2\pi\mathbb{Z} \\
\sigma_{\text{ess}}(A) &= \{\lambda \in \mathbb{C} \mid \lambda I - A \text{ is not semi-Fredholm}\} = \varnothing
\end{aligned}$$

The quadratic form $a(\cdot,\cdot)$ is positive definite on the space

$$\left\{ x^+(t) = \sum_{k>0} e^{2\pi k t J_0}\hat{x}_k \;\middle|\; \hat{x} \in \mathbb{R}^{2n}, t \in S^1 \right\},$$

and is negative definite on the space

$$\left\{ x^-(t) = \sum_{k<0} e^{2\pi k t J_0}\hat{x}_k \;\middle|\; \hat{x} \in \mathbb{R}^{2n}, t \in S^1 \right\}.$$

Both of these spaces are infinite dimensional. The second differential of $\mathcal{A}_H$ at x takes the form

$$\begin{aligned}
d^2\mathcal{A}_H(x)(\xi,\xi) &= -\int_{S^1} \langle J_0\dot{\xi}(t), \xi(t)\rangle - \int_{S^1} d^2H(t, x(t))(\xi,\xi)dt \\
&= 2\pi \sum_{k\in\mathbb{Z}} k\|\hat{\xi}_k\|^2 - d^2b(x)(\xi,\xi) \qquad \forall \xi \in \Gamma(x^*T\mathbb{R}^{2n}).
\end{aligned}$$

It follows that $d^2\mathcal{A}_H(x)$ is also positive definite on an infinite dimensional space and negative definite on an infinite dimensional space. Hence, all critical points of $\mathcal{A}_H$ have infinite Morse index and co-index, so the classical extension of Morse theory to infinite dimensional manifolds does not give any information: passing a critical value would change the sublevels of $\mathcal{A}_H$ by attaching an infinite dimensional cell along its boundary, and since infinite dimensional sphere are contractible, the homotopy type of the sublevels never changes.

1.3 Fractional Sobolev spaces and regularized gradient

It is a general fact in nonlinear analysis that the "right" functional space for studying a functional by topological methods is the space with the weakest topology on which this functional is "regular", for instance continuously differentiable. Indeed, regularity is required to obtain a negative gradient equation providing a well-posed ODE on the infinite dimensional space. The choice of a finer topology keeps the regularity (sometimes even improves it), however usually causes the failure of an important compactness condition known as the Palais–Smale condition. Observe that

$$\|\hat{x}_0\|^2 + 2\pi \sum_{k\in\mathbb{Z}} |k| \|\hat{x}_k\|^2$$

is the square of the $H^{\frac{1}{2}}$-norm of $x : S^1 \to \mathbb{R}^{2n}$. In general, for $s \in \mathbb{R}_{\geq 0}$, the fractional Sobolev space

$$H^s(S^1, \mathbb{R}^{2n}) := \left\{ x \in L^2(S^1, \mathbb{R}^{2n}) \;\middle|\; \sum_{k\in\mathbb{Z}} |k|^{2s} \|\hat{x}_k\|^2 \right\}$$

is a Hilbert space with respect to the H^s-inner product

$$\langle x, y \rangle_{H^s} = \langle \hat{x}_0, \hat{y}_0 \rangle + 2\pi \sum_{k\in\mathbb{Z}\setminus\{0\}} |k|^{2s} \langle \hat{x}_k, \hat{y}_k \rangle.$$

We have the following Sobolev embeddings.

Theorem 1.3.1 ([Abb01, Theorem 3.1.1])**.** *The following embeddings are compact.*

(i) $H^s(S^1, \mathbb{R}^{2n}) \hookrightarrow H^r(S^1, \mathbb{R}^{2n})$ *if* $s > r > 0$*;*

(ii) $H^s(S^1, \mathbb{R}^{2n}) \hookrightarrow C^q(S^1, \mathbb{R}^{2n})$ *if* $s > \frac{1}{2} + q$ *and* $q \in \mathbb{Z}_{\geq 0}$*;*

(iii) $H^{\frac{1}{2}}(S^1, \mathbb{R}^{2n}) \hookrightarrow L^p(S^1, \mathbb{R}^{2n})$ $\forall p \in [1, \infty)$.

Notice that functions in $H^{\frac{1}{2}}$ need not be continuous, and not even need to lie in L^∞. The compact inclusion

$$j : H^{\frac{1}{2}}(S^1, \mathbb{R}^{2n}) \hookrightarrow L^2(S^1, \mathbb{R}^{2n})$$

has adjoint

$$j^* : L^2(S^1, \mathbb{R}^{2n}) \to H^{\frac{1}{2}}(S^1, \mathbb{R}^{2n})$$

defined via

$$\langle j(x), y \rangle_{L^2} = \langle x, j^*(y) \rangle_{H^{\frac{1}{2}}}$$

which is also compact. It actually factors through

$$j^* : L^2(S^1, \mathbb{R}^{2n}) \to H^1(S^1, \mathbb{R}^{2n}) \to H^{\frac{1}{2}}(S^1, \mathbb{R}^{2n})$$

such that

$$j^*(y) = y_0 + \sum_{k \in \mathbb{Z}\setminus\{0\}} \frac{1}{2\pi|k|} e^{2\pi k t J_0} y_k.$$

The leading term of the regularized action functional (4) under $H^{\frac{1}{2}}$-norm is

$$a(x) = \frac{1}{2}\|x^+\|^2_{H^{\frac{1}{2}}} - \frac{1}{2}\|x^-\|^2_{H^{\frac{1}{2}}}.$$

The term (5) is not well-defined in the entire $H^{\frac{1}{2}}$, since $x \in H^{\frac{1}{2}}$ is not continuous and hence bounded in general, thus $H(t, x(t))$ needs not be integrable. The Hamiltonian H is said to be satisfying the following **polynomial growth condition**: $\exists C > 0, p > 0$ s.t.

$$\|\nabla^2 H(t, x)\| \leq C(1 + \|x\|^p). \tag{7}$$

This also implies polynomial growth of H and ∇H. Then under this assumption (5) is well-defined in $H^{\frac{1}{2}}(S^1, \mathbb{R}^{2n})$. We assume (7) whenever we need the action functional to be defined on $H^{\frac{1}{2}}(S^1, \mathbb{R}^{2n})$. Then the Sobolev embedding (iii) in Theorem 1.3.1 guarantees that $\mathcal{A}_H$ is C^2 on the entire $H^{\frac{1}{2}}(S^1, \mathbb{R}^{2n})$. In comparison, when H satisfies only (3), the action functional $\mathcal{A}_H$ is merely C^1 on $H^{\frac{1}{2}}(S^1, \mathbb{R}^{2n})$. The Sobolev space

$$E := H^{\frac{1}{2}}(S^1, \mathbb{R}^{2n})$$

admits decomposition w.r.t. the Fourier modes:

$$\begin{aligned}
E &= E^0 \oplus E^+ \oplus E^- \\
E^0 &:= \{x_0 \in \mathbb{R}^{2n}\} = \mathbb{R}^{2n} \\
E^+ &:= \left\{ x^+(t) = \sum_{k>0} e^{2\pi k t J_0} \hat{x}_k \;\middle|\; \hat{x} \in \mathbb{R}^{2n}, t \in S^1 \right\} \\
E^- &:= \left\{ x^-(t) = \sum_{k<0} e^{2\pi k t J_0} \hat{x}_k \;\middle|\; \hat{x} \in \mathbb{R}^{2n}, t \in S^1 \right\}
\end{aligned}$$

along with the following projection operators:

$$\begin{aligned}
P^0 &: E \to E^0 \\
P^+ &: E \to E^+ \\
P^- &: E \to E^-
\end{aligned}$$

The action functional can be represented by

$$\mathcal{A}_H(x) = \frac{1}{2}(\|P^+(x)\|_{H^{\frac{1}{2}}} - \|P^-(x)\|_{H^{\frac{1}{2}}}) - b(x). \tag{8}$$

The differential of action functional with respect to the $H^{\frac{1}{2}}$-inner product

$$d\mathcal{A}_H(x)(y) = \langle (P^+ - P^-)(x), y\rangle_{H^{\frac{1}{2}}} - \langle j^*\nabla_{L^2} b(x), y\rangle_{H^{\frac{1}{2}}}$$

gives rise to the $H^{\frac{1}{2}}$-gradient of $\mathcal{A}_H$

$$\nabla_{H^{\frac{1}{2}}}\mathcal{A}_H(x) = P^+(x) - P^-(x) - j^*\nabla_{L^2} b(x)$$

where $\nabla_{L^2} b(x) = \nabla H(t,x) \in L^2$, which is not necessarily compact in general, and becomes compact under compact map j^* to $H^{\frac{1}{2}}$. The $H^{\frac{1}{2}}$-Hessian then reads

$$\nabla^2_{H^{\frac{1}{2}}}\mathcal{A}_H(x)(\cdot,\cdot) = \langle (P^+ - P^-)(\cdot), \cdot\rangle_{H^{\frac{1}{2}}} - j^*\nabla^2 H(t,x)(\cdot,\cdot).$$

We can view

$$P^+ - P^- = j^*A =: S$$

as invertible self-adjoint operator on E.

1.4 Hamiltonian system in T^{2n} and its functional setup

Despite the infiniteness of the Morse indices, the study of the action functional on loops in a more general symplectic manifold (M,ω) has the extra difficulty that the Sobolev class $H^{\frac{1}{2}}(S^1, M)$ taking values in a manifold do not form a Hilbert manifold. Indeed, in general the space of Sobolev maps between manifolds has a good differentiable structure only when the Sobolev space embeds into the space of continuous loops, and the Sobolev space $H^{\frac{1}{2}}(S^1, M)$ does not have this property (the exponent $\frac{1}{2}$ is critical for the embedding into the space of continuous functions). We consider $M = T^{2n}$ with ω_0 descending from $\mathbb{R}^{2n}$ under the quotient $\mathbb{R}^{2n}/\mathbb{Z}^{2n}$. A contractible loop x on T^{2n} lifts to a closed loop on $\mathbb{R}^{2n}$ unique up to integer translations in $\mathbb{Z}^{2n}$, hence it can be written as

$$x(t) = \overline{x}_0 + \sum_{k\in\mathbb{Z}\setminus\{0\}} e^{2\pi k t J_0}\hat{x}_k, \qquad \overline{x}_0 \in T^{2n}, \hat{x}_k \in \mathbb{R}^{2n},$$

The space of contractible smooth loops $C^\infty_{\text{contr.}}(S^1, T^{2n})$ can be identified with the product space

$$C^\infty_{\text{contr.}}(S^1, T^{2n}) \cong T^{2n} \times F^\infty, \qquad F^\infty := \left\{ x \in C^\infty(S^1, \mathbb{R}^{2n}) \;\middle|\; \int_{S^1} x(t)dt = 0 \right\},$$

which is the quotient under $\mathbb{Z}^{2n}$-action of the space

$$\mathbb{R}^{2n} \times F^\infty \cong C^\infty(S^1, \mathbb{R}^{2n}).$$

In the $H^{\frac{1}{2}}$-setting, we have

$$\Lambda := H^{\frac{1}{2}}(S^1, T^{2n}) \cong T^{2n} \times F, \qquad F = \left\{ x \in E \;\middle|\; \int_{S^1} x(t)dt = 0 \right\},$$

where $E := H^{\frac{1}{2}}(S^1, \mathbb{R}^{2n})$ and actually

$$F = E^+ \oplus E^- = (F \cap E^+) \oplus (F \cap E^-) = F^+ \oplus F^-.$$

This implies that Λ is a Hilbert manifold with trivial tangent bundle

$$T\Lambda = \Lambda \times E$$

regarding the definition.

Clearly $\mathcal{A}_H$ is invariant under the $\mathbb{Z}^{2n}$-action and hence descends to T^{2n}. The Hamiltonian $H(t,x) \in C^2(S^1 \times T^{2n})$ is automatically bounded on $x \in H^{\frac{1}{2}}(S^1, T^{2n})$ since the image of the loop $x(t) \in T^{2n}$ is bounded due to the compactness of T^{2n}, thus the term (5) is well-defined without any assumption on the Hamiltonian H.

2 Morse theory for Hamiltonian systems

In the space $E := H^{\frac{1}{2}}(S^1, \mathbb{R}^{2n})$, the negative gradient flow equation

$$\dot{x}(t) = -\nabla_{H^{\frac{1}{2}}} \mathcal{A}_H(x) \tag{9}$$

is well-posed for initial value problem, and its solution can be expressed by

$$x(s) = x^0 + e^{-s}x^+ + e^s x^- - \int_0^s (P^0 + e^{-s+\tau}P^+ + e^{s-\tau}P^-) j^* \nabla H(t, x(\tau)) d\tau \qquad \forall s \in \mathbb{R}.$$

Moreover, considering bounded negative gradient flow trajectories

$$\sup_{s \in \mathbb{R}} |\mathcal{A}_H(x(s))| < \infty$$

which implies

$$\int_{-\infty}^{+\infty} \|\nabla_{H^{\frac{1}{2}}} \mathcal{A}_H\|^2_{H^{\frac{1}{2}}} ds < \infty,$$

then we can find a sequence $s_n \to \infty$ s.t. $\nabla_{H^{\frac{1}{2}}} \mathcal{A}_H(x(s_n)) \to 0$ and $\mathcal{A}_H(x(s_n))$ converges to some real number. The careful choice of the asymptotic behavior of the Hamiltonian function H, e.g. X_H has no 1-periodic orbits in the region of $\mathbb{R}^{2n}$ where $\|x\|$ is large. From this dynamical behavior, the action functional $\mathcal{A}_H$ satisfies the Palais-Smale condition **(PS)**. This is a strong compactness condition guaranteeing a critical point in many variational problems. Therefore, a subsequence of $\{x(s_n)\}_{n\in\mathbb{N}}$ converges in E to a critical point of $\mathcal{A}_H$. Moreover, the set $\mathrm{Crit}(\mathcal{A}_H)$ of all critical points of $\mathcal{A}_H$ is compact and one verifies readily that

$$\lim_{s\to\pm\infty} x(s) \in \mathrm{Crit}(\mathcal{A}_H)$$

for every bounded trajectories. Thes are the so-called connecting trajectories. Actually, the negative gradient flow trajectory converging to a critical point of $\mathcal{A}_H$ exponentially. In the space $H^1(S^1, \mathbb{R}^{2n})$, which is embedded into $C^0(S^1, \mathbb{R}^{2n})$, the negative gradient flow equation (9) is locally well-posed but has no exponential converging features towards crtical points.

2.1 Palais-Smale condition

Definition 2.1.1 (Palais-Smale)**.** Given a Riemannian Hilbert manifold $(\mathfrak{M}, \langle\cdot,\cdot\rangle)$, and $f \in C^1(\mathfrak{M}, \overline{\mathbb{R}})$[1] with gradient ∇f w.r.t. $\langle\cdot,\cdot\rangle$. f satisfies **(PS)** condition on $I \subseteq \overline{\mathbb{R}}$ if for any sequence $\{x_n\}_{n\in\mathbb{N}} \subset \mathfrak{M}$ with strong convergences $f(x_n) \xrightarrow{n\to\infty} c$ for some $c \in I \subseteq \overline{\mathbb{R}}$ and $df(x_n) \xrightarrow{n\to\infty} 0$ or $\nabla f(x_n) \xrightarrow{n\to\infty} 0$ has convergent subsequence in $\mathfrak{M}$.

Since $\overline{\mathbb{R}}$ is compact, $\mathrm{Crit}(f)$ is compact when f satisfies **(PS)** on $\overline{\mathbb{R}}$.

Lemma 2.1.2 ([Cha93, Lemma 3.2])**.** *If $f \in C^1(\mathfrak{M}, \overline{\mathbb{R}})$ satisfies* **(PS)** *on $[a,b]$, and $\mathrm{Crit}(f) \cap f^{-1}([a,b]) = \varnothing$, then $f^{-1}((-\infty,b])$ strong deformation retracts to $f^{-1}((-\infty,a])$.*

Theorem 2.1.3 ([Cha93, Theorem 4.3])**.** *Let $f \in C^2(\mathfrak{M}, \overline{\mathbb{R}})$ and $a < b$ be regular values of f. Suppose f satisfies* **(PS)** *on $[a,b]$ and all its x critical points in $f^{-1}([a,b])$ are non-degenerate (hence their Morse indices $m(x)$ are finite). Then there exists a polynomial $Q(t)$ with $\mathbb{Z}_{\geq 0}$-coefficient s.t. the following identities holds:*

$$\sum_{x\in\mathrm{Crit}(f)\cap f^{-1}([a,b])} t^{m(x)} = P(t, f^{-1}((-\infty,b]), f^{-1}((-\infty,a])) + (1+t)Q(t)$$

[1] $\overline{\mathbb{R}} = \mathbb{R} \cup \{\pm\infty\}$

where P is the Poincaré polynomial of pairs:

$$P(t, f^{-1}((-\infty, b]), f^{-1}((-\infty, a])) = \sum_{k \in \mathbb{Z}_{\geq 0}} \operatorname{rank} H_k(f^{-1}((-\infty, b]), f^{-1}((-\infty, a]))t^k$$

In above, we use convention $t^\infty = 0$ relating to the fact that critical points with infinite Morse indices do not change the homotopy type of sublevel sets. As a consequence, we have

$$\sum_{j=0}^{k}(-1)^{k-j}M_j(a,b) \geq \sum_{j=0}^{k}(-1)^{k-j}b_j(a,b)$$
$$\sum_{k=0}^{\infty}(-1)^{k}M_k(a,b) \geq \sum_{k=0}^{\infty}(-1)^{k}b_k(a,b)$$

where

$$M_k(a,b) = \#\{x \in \operatorname{Crit}(f) \cap f^{-1}([a,b]) : m(x) = k\}$$
$$b_k(a,b) = \operatorname{rank} H_k(f^{-1}((-\infty, b]), f^{-1}((-\infty, a]))$$

are the Morse type numbers and Betti numbers.

When dealing with strongly indefinite functionals

$$f(x) = \frac{1}{2}\langle S_0 x, x\rangle_E + b(x)$$

with S_0 self-adjoint invertible and $\nabla_E b : E \to E$ compact, it is often convenient to restrict to a finite dimensional subspace to obtain the existence of constrained critical points, then attempt to take a limit by choosing larger and larger subspaces. A stronger version of (PS) condition is needed for such approach. Assume $f \in C^1(E)$ where E is a separable Hilbert space. Let (P_n) be a sequence of orthogonal projections with finite ranks s.t. the **Galerkin's finite approximation** assumptions w.r.t. S_0 are satisfied:

(a) $E_n := \operatorname{im}(P_n)$, then $E_n \subset E_m$ if $n \leq m$.

(b) $P_n \xrightarrow{n\to\infty} I$ strongly.

(c) $[S_0, P_0] := S_0P_n - P_nS_0 \xrightarrow{n\to\infty} 0$ in operator norm.

In fact, this always exists. See [Abb01, Proposition 2.5.1].

Definition 2.1.4. The functional f satisfies **(PS)*** on $\overline{\mathbb{R}}$ w.r.t. (P_n) if every sequence $\{x_n\}_{n\in\mathbb{N}}$ s.t. $x_n \in E_n$, $f(x_n) \xrightarrow{n\to\infty} c$ for $c \in \overline{\mathbb{R}}$ and $P_n(\nabla_E f(x_n)) \xrightarrow{n\to\infty} 0$, has convergent subsequence. Such sequence is called **(PS)***-sequence.

Denote $f_n := f|_{E_n}$, the assumption **(a)** on (P_n) guarantees that if f satisfies **(PS)*** then each f_n satisfies **(PS)**. Furthermore, every accumulation point of a **(PS)***-sequence is a critical point of f. The **(PS)*** condition turns out to be very useful when dealing with the functional of the form

$$f(x) = \frac{1}{2}\langle S_0 x, x\rangle_E + b(x) \tag{10}$$

where S_0 is an invertible slef-adjoint operator and b has compact gradient. Under the Galerkin's finite approximation assumptions **(a)**,**(b)**,**(c)**, the functional of the form (10) has the following property: every bounded **(PS)***-sequence has convergent subseqquence. In fact, if (x_n) is a bounded **(PS)***-sequence, then up to subsequence we can assume that $\nabla_E b(x_n) \xrightarrow{n\to\infty} y$, thus

$$S_0 x_n = P_n S_0 + [S_0, P_n]x_n = P_n \nabla_E f(x_n) - P_n \nabla_E b(x_n) + [S_0, P_n]x_n \xrightarrow{n\to\infty} -y.$$

Since S_0 is invertible, $x_n \xrightarrow{n\to\infty} -S_0^{-1}y$. Similar argument shows that when f is of the form (10), then every bounded **(PS)**-sequence has convergent subsequence.

Lemma 2.1.5 ([Abb01, Lemma 5.1.2]). *Assume f on separable Hilbert space is of* (10) *and satisfies*

$$\|\nabla_E b(x)\|_E = o(\|x\|_E), \qquad \|x\|_E \to \infty. \tag{11}$$

Then f satisfies **(PS)*** *on $\overline{\mathbb{R}}$ w.r.t. (P_n).*

Proof. Let $(x_n) \subset E$ be such that $x_n \in E_n$ and $P_n \nabla_E f(x_n) \xrightarrow{n\to\infty} 0$. By (11), there exists $c > 0$ s.t.

$$\|\nabla_E b(x)\|_E \le \frac{\|x\|_E}{2\|S_0^{-1}\|_{\mathcal{L}(E)}} + c$$

For n large, we have

$$\begin{aligned} 1 &\ge \|P_n \nabla_E f(x_n)\|_E \\ &= \|S_0 x_n + [P_n, S_0]x_n + P_n \nabla_E b(x_n)\|_E \\ &\ge \frac{\|x_n\|_E}{2\|S_0^{-1}\|_{\mathcal{L}(E)}} - \|[P_n, S_0]\|_{\mathcal{L}(E)}\|x_n\|_E - c \end{aligned}$$

Since $[P_n, S_0] \xrightarrow{n\to\infty} 0$, the above inequality implies that (x_n) is bounded. As f consists of the sum of a non-degenerate quadratic form and a term with compact gradient, (x_n) must have convergent subsequence. □

Remark 2.1.6. The **(PS)*** condition ensures that every sequence of critical points of (f_n) converges, up to subsequence, to some critical point of f.

In particular, for the Hamiltonian system on $\mathbb{R}^{2n}$:

Lemma 2.1.7 ([BSW08, Lemma 6.4])**.** *Suppose $H \in C^\infty(S^1, \mathbb{R}^{2n})$ satisfies*

- $H(t, 0) \equiv 0$,
- *H satisfies (3), i.e. asymptotic linear.*

If $\sigma(J_0 B_\infty) \cap i\mathbb{Z} = \varnothing$, then the functional $\mathcal{A}_H$ satisfies **(PS)*** *condition w.r.t. (P_n).*

Lemma 2.1.8 ([Abb01, §4.2, Claim 1])**.** *When H satisfies (7), the action functional $\mathcal{A}_H$ satisfies the* **(PS)*** *condition w.r.t. (P_n).*

Remark 2.1.9. In above lemmas, we have $E := H^{\frac{1}{2}}(S^1, \mathbb{R}^{2n})$ and $P_n : E \to E_n$ where

$$E_n := \left\{ x_n(t) = \sum_{|k| \le n} e^{2\pi k t J_0} \hat{x}_k \;\middle|\; \hat{x} \in \mathbb{R}^{2n}, t \in S^1 \right\}.$$

For Hamiltonian system on T^{2n}, since $\nabla_{H^{\frac{1}{2}}} b$ is compact and bounded, then it is easy to show that $\mathcal{A}_H$ satisfies **(PS)*** on $\overline{\mathbb{R}}$: every sequence $(x_n) \subset \Lambda$ s.t. $x_n \in \Lambda_n$ and $P_n \nabla_{H^{\frac{1}{2}}} \mathcal{A}_H(x_n) \to 0$ has convergent subsequence.

2.2 Relative Morse index

In contrast to the case of a smooth Morse function on a finite-dimensional manifold, as it will turn out, we have to deal with the strongly indefinite functional $\mathcal{A}_H$ defined on the Hilbert manifold Λ, i.e. $\mathcal{A}_H$ possesses infinite Morse indices and coindices. Abbondandolo and Majer developed methods enabled to construct an infinite-dimensional Morse theory for such functionals with a weaker replacement for Morse index.

We consider the Hilbert Grassmannian $\mathrm{Gr}(E)$ of a separable Hilbert space E, which is the set of all closed subspaces of E. For $V \in \mathrm{Gr}(E)$, we denote by P_V the orthogonal projection onto V. The distance between two subspaces $V, W \in \mathrm{Gr}(E)$ is defined by

$$\mathrm{dist}(V, W) := \|P_V - P_W\|_E.$$

This makes $\mathrm{Gr}(E)$ become a complete metric space with connected components

$$\mathrm{Gr}_{n,m}(E) := \{V \in \mathrm{Gr}(E) : \dim V = n, \mathrm{codim}\, V = m\}$$

where $n, m \in \mathbb{N} \cup \{\infty\}$ and $n + m = \infty$.

Definition 2.2.1. A pair (V, W) of closed subspaces in E is called **Fredholm pair** if $V \cap W$ is finite dimensional and $V + W$ is closed and finite dimensional. In this case,

$$\mathrm{ind}(V, W) := \dim(V \cap W) - \mathrm{codim}(V + W)$$

is called the Fredholm index of (V, W).

Remark 2.2.2. An operator $F : X \to Y$ between Hilbert spaces is Fredholm if and only if $(\mathrm{graph}(F), X \times \{0\})$ is a Fredholm pair in $\mathrm{Gr}(X \times Y) \times \mathrm{Gr}(X \times Y)$. The set of all Fredholm pairs in E is an open subset of $\mathrm{Gr}(E) \times \mathrm{Gr}(E)$ and the Fredholm index is a continuous (i.e. locally constant) function on it.

Definition 2.2.3 ([Abb01, §2.2], [Hec13, Definition 2.11])**.** Let V, W be closed subspaces in E. Then V is called a **compact pertubation** of or **commmensurable** with W if $P_V - P_W$ is compact operator. In this case, $(V, W^\perp)$ is Fredholm pair and its index defines the **relative dimension** of V w.r.t. W, i.e.

$$\dim(V, W) := \mathrm{ind}(V, W^\perp) = \dim(V \cap W^\perp) - \dim(V^\perp \cap W)$$

Remark 2.2.4.

- The commensurability is an equivalence relation.
- The notions of commensurability and the relative dimension are independent on the equivalent inner product chosen on E. See [Abb01, Proposition 2.1.4]
- If (V, W) is a Fredholm pair, and U is a compact pertubation of V, then (U, V) is also a Fredholm pair, and

$$\mathrm{ind}(U, W) = \dim(U, V) + \mathrm{ind}(V, W).$$

Proposition 2.2.5 ([Abb01, Proposition 2.3.2], [Hec13, Proposition 2.12])**.** *Let S and T be two self-adjoint Fredholm operators on Hilbert space E with $S - T$ compact. Then the positive and negative eigenspaces of S are commensurable with the positive and negative eigenspaces of T respectively.*

Proof. For any polynomial $p \in \mathbb{R}[t]$, the operator $p(S) - p(T)$ is compact since it belongs to the two-sided ideal spanned by $S - T$ in $\mathcal{L}(E)$. By density, for any

continuous function g on $\sigma(S)\cup\sigma(T) \subset \mathbb{R}$, the image $g(\sigma(S))-g(\sigma(T))$ is compact. Since S, T are self-adjoint, 0 is not an accumulation point for $\sigma(S)\cup\sigma(T)$, hence the characteristic function $\chi_{\mathbb{R}_\pm}$ are continuous on $\sigma(S) \cup \sigma(T)$. By spectral theorem,

$$P_{E^\pm(S)} - P_{E^\pm(T)} = \chi_{\mathbb{R}_\pm}(\sigma(S)) - \chi_{\mathbb{R}_\pm}(\sigma(T))$$

are compact operators. □

Let S be a bounded self-adjoint operator which is Fredholm, the symmetric bilinear form

$$a(\cdot,\cdot) = \langle S\cdot,\cdot\rangle_E$$

is called a **Fredholm form**. This is equivalently that 0 is not an accumulating point for the spectrum of the bounded operator S and $\ker(S)$ is finite dimensional. This determine a unique S-invariant orthogonal splitting

$$E = E^+(S) \oplus E^-(S) \oplus \ker(S)$$

such that $a(\cdot,\cdot)$ is coercive on $E^+(S)$, i.e. $a(x,x) \geq c\|x\|_E^2$ for some $c > 0$, and $-a(\cdot,\cdot)$ is coercive on $E^-(S)$. The above splitting is also a-orthogonal.

Definition 2.2.6. Let $a(\cdot,\cdot) = \langle S\cdot,\cdot\rangle_E$ be a Fredholm form. The **nulllity** of a is

$$\nu(a) = \nu(S) := \dim\ker(S).$$

Assume that $E^-(S)$ is commensurable with some closed subspace V. The **relative Morse index** of a w.r.t. V is

$$i_V(a) = i_V(S) := \dim(E^-(S), V).$$

Remark 2.2.7. When $V = \{0\}$, the relative Morse index is just the ordinary Morse index $i(a) = i(S)$. Every quadratic form a perturbed by a compact operator will also be a Fredholm form, its negative eigenspace is still commensurable with V so that $i_V(a)$ is well-defined.

The functional (10) yields Fredholm form d^2f on E with Hessian at $x \in \mathrm{Crit}(f)$:

$$\nabla_E^2 f(x) = S_0 + \nabla_E^2 b(x)$$

Let $E = E^+(S_0) \oplus E^-(S_0)$ be orthogonal splitting into positive and negative eigenspaces of S_0. By Proposition 2.2.5, the negative eigenspace $E^-(\nabla_E^2 f)$ of $\nabla_E^2 f$ is commensurable with $E^-(S_0)$. Let $x \in \mathrm{Crit}(f)$ with f of the form (10). The relative Morse index of x w.r.t. $E^-(S_0)$ is

$$m_{E^-(S_0)}(x) := i_{E^-(S_0)}(d^2f(x)) = \dim(E^-(\nabla_E^2 f(x)), E^-(S_0)).$$

Theorem 2.2.8 ([Abb01, Theorem 5.1.1]). *Assume f on separable Hilbert space E of the form* (10) *satisfies* (11). *If the critical points of f are non-degenerate, then* $\mathrm{Crit}(f)$ *is a finite set and the following equality holds*

$$\sum_{x\in\mathrm{Crit}(f)} t^{m_{E^-(S_0)}(x)} = 1 + (1+t)Q(t)$$

where Q is a Laurent polynomial with $\mathbb{Z}_{\geq 0}$-coefficients.

The relative Morse index of self-adjoint operator S as perturbation of invertible self-adjoint operator S_0 by compact operator and its finite dimensional reductions P_nSP_n (still self-adjoint) where (P_n) is a sequence of finite dimensional approximation scheme w.r.t. S_0 has the following relation.

Theorem 2.2.9 ([Abb01, Theorem 2.5.4]). *Let S_0 be a self-adjoint invertible operator and (P_n) be an approximation scheme w.r.t. S_0. Let S be a self-adjoint operator s.t. $S - S_0$ is compact, and $P_{\ker(S)}$ be the orthogonal projection onto* $\ker(S)$. *Then*

$$i_{E^-(S_0)}(S) = \lim_{n\to\infty}[i(P_n(S + P_{\ker(S)})P_n) - i(P_nS_0P_n)].$$

The above limits are uniform, for S varying in a compact set.

A converse statement to Remark 2.1.6 holds: every critical point of f can be found as the limit of a sequence of critical points of (f_n).

Lemma 2.2.10 ([Abb01, Lemma 5.1.3]). *Assume f is of* (10). *Let $x \in \mathrm{Crit}(f)$ be non-degenerate. Then there exists $n_0 \in \mathbb{N}$ and $r > 0$ s.t. for every $n \geq n_0$ the set $E_n \cap B_r(x)$ contains a critical point $x_n \in \mathrm{Crit}(f_n)$. Moreover, $x_n \in \mathrm{Crit}(f_n)$ is non-degenerate, $x_n \xrightarrow{n\to\infty} x$ and*

$$m_{E^-(S_0)}(x; f) = m(x_n; f_n) - i(P_nS_0P_n) \qquad \forall n \geq n_0.$$

The idea is to use implicit function theorem to show $\{y \in E : P_n\nabla_E f(y) = 0\}$ intersects E_n transversally near $x \in \mathrm{Crit}(f)$. The unique point of intersection will be the required constrained critical point x_n.

For a non-linear periodic Hamiltonian system (1), denote for the linearized Hamiltonian system (2) the operator

$$B(t) := \nabla^2 H(t, x(t))$$

at a solution $x \in \mathrm{Per}(H)$. The nullity of x is defined to be

$$\nu(x) := \nu(B) = \nu(d^2\mathcal{A}_H(x)), \tag{12}$$

and the relative Morse index of x is

$$m_{E^-(S_0)}(x) := i_{E^-(S_0)}(d^2\mathcal{A}_H(x)), \tag{13}$$

where

$$\begin{aligned}
\mathcal{A}_H(x) &= \frac{1}{2}\langle S_0 x, x\rangle_E + b(x),\\
\langle S_0 x, x\rangle_E &= -\int_{S^1}\langle J_0\dot{x}(t), x(t)\rangle dt + \frac{1}{2}\left\|\int_{S^1} p(t)dt\right\|^2 - \frac{1}{2}\left\|\int_{S^1} q(t)dt\right\|^2\\
&= \|p_0\|^2 - \|q_0\|^2 + 2\pi\sum_{k\in\mathbb{Z}\setminus\{0\}} k\|\hat{x}_k\|^2,\\
S_0 &= p_0 - q_0 + \sum_{k\in\mathbb{Z}\setminus\{0\}}\frac{k}{|k|}e^{2\pi k t J_0}\hat{x}_k,\\
b(x) &= -\int_{S^1} H(t, x(t))dt - \frac{1}{2}\left\|\int_{S^1} p(t)dt\right\|^2 + \frac{1}{2}\left\|\int_{S^1} q(t)dt\right\|^2.
\end{aligned}$$

Then

$$E^-(S_0) = \mathbb{R}^n \times E^-.$$

2.3 Morse homology for action functional on T^{2n}

Let X be a C^1-vector field on a Hilbert manifold $\mathfrak{M}$. The set of singular points of X is denoted by $\mathrm{Sing}(X)$. Then $x \in \mathrm{Sing}(X)$ is called **hyperbolic** iff. for the Jacobian $DX(x) : T_x\mathfrak{M} \to T_x\mathfrak{M}$ its spectrum $\sigma(DX(x)) \cap i\mathbb{R} = \varnothing$. A **Morse vector field** is a vector field X whose singular points are all hyperbolic. If there is $f \in C^1(\mathfrak{M})$ s.t.

$$df(x)(X(x)) < 0 \qquad \forall x \in \mathfrak{M} \setminus \mathrm{Sing}(X),$$

then f is called a **Lyapunov function** for X. A Morse vector field X is called **gradient-like** if it has a non-degenerate Lyapunov function.

Proposition 2.3.1 ([Hec13, Proposition 2.6]). *There is a residual set $\mathcal{H}_{\mathrm{reg}} \subset C^\infty(S^1 \times T^{2n})$ of Hamiltonians s.t. $\nabla_{H^{\frac{1}{2}}}\mathcal{A}_H$ is a Morse vector field for every $H \in \mathcal{H}_{\mathrm{reg}}$. In particular, the set of all contractible periodic Hamiltonian orbits $\mathrm{Per}_0(H)$ is a finite set.*

Theorem 2.3.2 ([Hec13, Theorem 2.14]). *Let $x \in \mathfrak{M}$ be a hyperbolic singular point of a complete C^k-vector field X ($k \geq 1$) on a Hilbert manifold $\mathfrak{M}$. Then $W^u(x)$ and $W^s(x)$ are the images of injective C^k-immersions of $E^u_x := E^+(DX(x))$ and $E^s_x := E^-(DX(x))$ respectively. If in addition X possesses a Lyapunov function f non-degenerate at x, then these immersions are actually embeddings.*

Although in general $W^u(x)$ and $W^s(x)$ are infinite-dimensional, Abbondandolo and Majer formulated two conditions which guarantee that $W^u(x) \cap W^s(x)$ is a finite-dimensional manifold.

Definition 2.3.3 ([Hec13, Definition 2.15])**.** Let X be a C^1-Morse vector field on the Hilbert manifold $\Lambda = T^{2n} \times F$. If $V \subset F$ is a closed subspace and $\mathcal{V} = \Lambda \times V$ is a trivial subbundle of $T\Lambda$, then $\mathcal{V}$ is said to be admissible for X if

(C1) for every singular point $x \in \mathrm{Sing}(X)$, the unstable eigenspace E^u_x is commensurable with V;

(C2) for all $x \in \mathfrak{M}$ the operator $[DX(x), P_V]$ is compact.

The relative Morse index of $x \in \mathrm{Sing}(X)$ relative to $\mathcal{V}$ is defined by

$$m_{\mathcal{V}}(x) := \dim(E^u_x, V).$$

Remark 2.3.4. The subbundle $\mathcal{V}$ is φ_X-invariant, i.e.

$$D\varphi^s_X(x)(V) = V \qquad \forall (s,x) \in \mathbb{R} \times \Lambda$$

iff. $[DX(x), P_V] = 0$.

Lemma 2.3.5 ([Hec13, Lemma 2.16])**.** *Let $H \in \mathcal{H}_{\mathrm{reg}}$ be a generic Hamiltonian, and $X := -\nabla_{H^{\frac{1}{2}}} \mathcal{A}_H$ is thus a Morse vector field on Λ. Then the subbundle*

$$\Lambda \times (\mathbb{R}^n \times E^-) \subset T\Lambda$$

is admissible for X.

Proof. By assumption, X is Morse, the operator $DX(x)$ is hyperbolic for all $x \in \mathrm{Sing}(X)$. Hence

$$\sigma(DX(x)) = \sigma^+(DX(x)) \sqcup \sigma^-(DX(x)).$$

Furthermore $DX(p)$ satisfies the estimate for all $\xi \in F$:

$$\begin{aligned} \|\xi\|_{Hfrac12} &= \|P^0\xi + DX(x)\xi + j^*\nabla^2 H(t,x)\xi\|_{H^{\frac{1}{2}}} \\ &\leq \|DX(x)\xi\|_{H^{\frac{1}{2}}} + \|j^*P^0\xi + j^*\nabla^2 H(t,x)\|_{H^{\frac{1}{2}}} \\ &\leq \|DX(x)\xi\|_{H^{\frac{1}{2}}} + (1 + \|H\|_{C^2})\|\xi\|_{L^2} \end{aligned}$$

Since $j^* : H^{\frac{1}{2}}(S^1, \mathbb{R}^{2n}) \to L^2(S^1, \mathbb{R}^{2n})$ is compact, the above estimate implies that $DX(x)$ is a semi-Fredholm operator. Since $DX(x)$ is also self-adjoint, its kernel

and cokernel coincide, and therefore $DX(x)$ is indeed Fredholm of index 0. In particular, 0 is not accumulation point in $\sigma(DX(x))$. We have

$$[DX(x), P^+] = j^*\nabla^2 H(t,x)P^+ - P^+ j^*\nabla^2 H(t,x)$$

is compact, which verifies (C2). Applying Proposition 2.2.5 to $S = DX(x)$ and $T = P^+ - P^-$, we know that E^u_x is commensurable with E^-, and therefore with $\mathbb{R}^n \times E^-$, hence verifying (C1). □

Proposition 2.3.6 ([Hec13, Proposition 2.17])**.** *Assume that X is C^1-Morse vector field defined on $\Lambda = T^{2n} \times F$, possessing a Lyapunov function f and an admissible trivial bundle $\mathcal{V} = \Lambda \times V \subset \Lambda \times F$. Then for every $x \in \mathrm{Sing}(X)$, the following holds:*

(i) for every $p \in W^u(x)$, then $T_pW^u(x)$ is commensurable with V and

$$\dim(T_pW^u(x), V) = m_{\mathcal{V}}(x);$$

(ii) for every $p \in W^s(x)$, then $(T_pW^s(x), V)$ is a Fredholm pair and

$$\mathrm{ind}(T_pW^s(x), V) = -m_{\mathcal{V}}(x);$$

(iii) let $x, y \in \mathrm{Sing}(X)$ and assume $W^u(x) \pitchfork W^s(y)$. Then $W^u(x) \cap W^s(y)$ is a submanifold of dimension

$$\dim(W^u(x) \cap W^s(y)) = m_{\mathcal{V}}(x) - m_{\mathcal{V}}(y).$$

Definition 2.3.7. Let X be a Morse vector field on the Hilbert manifold $\mathfrak{M}$ possessing a Lyapunov function f and an admissible trivial subbundle $\mathcal{V}$. We say that X satisfies the **Morse–Smale condition** up to order $k \in \mathbb{N}$ if for every pair of singular points $x, y \in \mathrm{Sing}(X)$ with $m_{\mathcal{V}}(x) - m_{\mathcal{V}}(y) \leq k$, and $W^u(x) \pitchfork W^s(y)$.

Abbondandolo and Majer established a transversality result by perturbing the metric on the Hilbert manifold. On the T^{2n} case, Hecht gives an alternative result by perturbing the negative gradient vector field directly by adding a small compact vector field $K : \Lambda \to F$. He shows that there is a generic compact perturbation K s.t. $X_K = -\nabla_{H^{\frac{1}{2}}}\mathcal{A}_H + K$ fulfills the Morse-Smale condition up to order 2. C.f. [Hec13, Theorem 2.20]. The perturbed negative gradient is a C^3-Morse vector field. The trivial subbundle $\mathcal{V} = \Lambda \times (\mathbb{R}^n \times E^+)$ is admissible for X_K. By counting rigid perturbed negative gradient flow trajectories, it turns out that the Morse complex

$$CM_*(\mathcal{A}_H) := \bigoplus_{x \in \mathrm{Crit}(\mathcal{A}_H)} \mathbb{Z}_2\langle x \rangle$$
$$\partial : CM_*(\mathcal{A}_H) \to CM_{*-1}(\mathcal{A}_H)$$

is well-defined, and the invariance of Morse homology holds:

$$HM_*(\mathcal{A}_H, X_K) \cong HM_*(\mathcal{A}_H, X_{K'})$$

for K, K' regular s.t. $X_K, X_{K'}$ satisfy (C1) and (C2) w.r.t. the same subbundle $\mathcal{V} = \Lambda \times (\mathbb{R}^n \times E^+)$. The main result in [Hec13] states:

$$HM_*(\mathcal{A}_H) \cong HF_*(H)$$

3 Conley index pair

3.1 Index pair

Let $\varphi_t : M \to M$ be a flow on a locally compact metric space. A subset $S \subseteq M$ is called an **invariant subset** iff. $\varphi_t(S) = S \ \forall t \in \mathbb{R}$. Given $N \subseteq M$, we consider the **maximal invariant subset**

$$I(N) = \bigcap_{t \in \mathbb{R}} \varphi_t(N)$$

A compact invariant set S is **isolated** iff. there is a compact neighborhood N of S s.t. $I(N) = S$. Such a neighborhood N is called an **isolated neighborhood**.

Definition 3.1.1. An **index pair** (N, L) for an isolated compact invariant subset S is a pair of compact sets $L \subset N$ s.t.

(i) $S = I(\overline{N \setminus L}) \subseteq \text{int}(N \setminus L)$.

(ii) L is **positively invariant** in N, i.e. $x \in L$ and $\varphi_{[0,t]}(x) \subseteq N \Rightarrow \varphi_{[0,t]}(x) \subseteq L$.

(iii) L is an **exit set** for N meaning each orbit that leaves N must go through L first. I.e. if $x \in N$ and $\varphi_t(x) \notin N$ for $t \in \mathbb{R}_+$, then there exists $t' \in [0, t]$ s.t. $\varphi_{[0,t']}(x) \subseteq N$ and $\varphi_{t'}(x) \in L$. Thus .

Theorem 3.1.2 ([BH04, Theorem 7.14])**.** *Every isolated compact invariant set admits an index.*

Theorem 3.1.3 ([BH04, Theorem 7.15], [CZ84, Theorem 3.2])**.** *If S is an isolated compact invariant set, (N, L) and (N', L') are two index pair for S, then N/L and N'/L' are homotopy equivalent as pointed spaces via maps that are induced by the flow.*

Definition 3.1.4. The **Conley index** of an isolated compact invariant set S is the homotopy type of N/L where (N, L) is an index pair for S, i.e. the unique pointed homotopy class $h(S) := [(N/L, *)]$ associated to S.

An index pair (N, L) is **regular** iff. the inclusion map $L \subset N$ is a cofibration. In fact, any index pair can be modified to a regular one.

3.2 Morse decomposition

Definition 3.2.1 ([CZ84, Definition 3.1]). A **Morse decomposition** of a compact invariant set S is a finite collection $\{M_i\}$ of disjoint compact invariant subsets in S which can be ordered $(M_1, M_2, \ldots, M_n)$ so that for every $x \in S \setminus \bigcup_{1\le j\le n} M_j$ there are $i < j$ s.t. $\omega^+(x) \subset M_i$ and $\omega^-(x) \subset M_j$. Such an ordering is called **admissible**. The elements of M_j of Morse decomposition of I is called **Morse sets** of S

There could be several admissible orderings of the same decomposition. For an admissible ordering $(M_1, \ldots, M_n)$, we define the subsets

$$M_{ji} = \{x \in S : \omega^+(x) \cup \omega^-(x) \subset M_i \cup M_{i+1} \cup \cdots \cup M_j\}$$

where $\omega^+(x) = I(\overline{\varphi_{[0,+\infty)}(x)})$ and $\omega^-(x) = I(\overline{\varphi_{(-\infty,0]}(x)})$. In particular, $M_{jj} = M_j$.

Proposition 3.2.2 ([CZ84, Proposition 3.1]). Assume $(M_1, \ldots, M_n)$ is an admissible ordering of a Morse decomposition of S. If $i \le j$, then

$$(M_1, \ldots, M_{i-1}, M_{ji}, M_{j+1}, \ldots, M_n)$$

is an admissible ordering of a Morse decomposition of S. Moreover,

$$(M_i, M_{i+1}, \ldots, M_{j-1}, M_j)$$

is an admissible ordering of a Morse decomposition of M_{ji}.

Proposition 3.2.3 ([CZ84, Proposition 3.2]). Assume S to be an isolated invariant set of a (local) flow and let $\{M_i\}$ be a Morse decomposition of S. Then the sets M_i are also isolated invariant sets of the (local) flow.

The following is the existence of filtration of index pairs.

Theorem 3.2.4 ([CZ84, Theorem 3.1]). Let S be an isolated invariant set and let $(M_1, \ldots, M_n)$ be an admissible ordering of a Morse decomposition of S. Then there exists an increasing sequence of compact sets

$$N_0 \subset N_1 \subset \cdots \subset N_n$$

s.t. for any $i \le j$ the pair (N_j, N_{i-1}) is an index pair for M_{ji}. In particular, (N_n, N_0) is an index pair for S and (N_j, N_{j-1}) is an index pair for M_j, Moreover, given any isolating neighborhood N of S and given any neighborhood U of S, the sets N_j can be chosen so that $\overline{N_n \setminus N_0} \subset U$, the sets N_j being positively invariant relative to N.

3.3 Morse relations

Proposition 3.3.1 ([CZ84, Porposition 3.3]). *Let $N_0 \subset N_1 \subset \cdots \subset N_n$ be compact spaces. Then*

$$\sum_{j=1}^{n} p(t, N_j, N_{j-1}) = p(t, N_n, N_0) + (1+t)Q(t)$$

where

$$Q(t) = \sum_{j=2}^{n} q(t, N_j, N_{j-1}, N_0).$$

Given a isolated invariant set S and a index pair (N, L) for it. Put

$$p(t, h(S)) := p(t, N, L).$$

Theorem 3.3.2 ([CZ84, Theorem 3.3]). Let $(M_1, \ldots, M_n)$ be an admissible ordering of a Morse decomposition of an isolated invariant set S. Then

$$\sum_{j=1}^{n} p(t, h(M_j)) = p(t, h(S)) + (1+t)Q(t)$$

where $Q(t)$ is a formal power series with $\mathbb{Z}_{\geq 0}$-coefficients.

Proof. By Theorem 3.2.4, there is a filtration $N_0 \subset N_1 \subset \cdots \subset N_n$ for the Morse decomposition s.t. (N_n, N_0) is an index pair for S and (N_j, N_{j-1}) is an index pair for M_j with $1 \leq j \leq n$. Then the statement follows immediately from Preposition 3.3.1. □

As a consequence, by the filtration of negative gradient flow of Morse function w.r.t. the Morse decomposition whose Morse sets are exactly the critical points, see [BH04, Example 7.17, §7.4], we have the following Morse relations.

Theorem 3.3.3 ([Abb01, Theorem 4.1.3]). Let $f \in C^2(\mathfrak{M})$ and suppose (N, L), where $f(N \setminus L)$ is bounded, is an index pair for the isolated invariant set $S := I(\overline{N \setminus L})$ w.r.t. $\Phi_t^{-\nabla f}$. Assume that f satisfies **(PS)** on $\overline{f(N \setminus L)}$ and all critical points in $N \setminus L$ are non-degenerate. Then

$$\sum_{x \in \mathrm{Crit}(f) \cap (N \setminus L)} t^{m(x)} = P(t, N, L) + (1+t)Q(t)$$

where $Q(t)$ is a formal power series with $\mathbb{Z}_{\geq 0}$-coefficients.

Proof of Theorem 2.2.8. Since the critical points are assumed to be non-degenerate, the fact that f satisfies **(PS)*** on $\overline{\mathbb{R}}$ by Lemma 2.1.5 implies that $\mathrm{Crit}(f)$ is a finite set. Then, by Lemma 2.2.10 :

$(*)$ For n sufficiently large , all critical points of f_n are non-degenerate and there is a bijection

$$j_n : \mathrm{Crit}(f) \to \mathrm{Crit}(f_n)$$

s.t.

$$m_{E^-(S_0)}(x; f) = m(j_n(x); f_n) - i(P_n S_0 P_n).$$

Moreover, we may assume:

$(**)$ For n sufficiently large,

$$\|(P_n S_0 P_n)^{-1}\|_{\mathcal{L}(E_n)} \leq 2\|S_0^{-1}\|_{\mathcal{L}(E)}.$$

This follows from the estimate

$$\|P_n S_0 P_n x\|_E \geq \left(\frac{1}{\|S_0^{-1}\|_{\mathcal{L}(E)}} - \|[P_n, S_0]\|_{\mathcal{L}(E)} \right) \|P_n x\|_E$$

and the fact $[P_n, S_0] \xrightarrow{n\to\infty} 0$ strongly.

It remains merely to show the Morse relations for f_n. By $(**)$, the operator $P_n S_0 P_n$ is invertible on E_n, denote by E_n^+ and E_n^- its positive and negative eigenspaces, P_n^+ and P_n^- the corresponding orthogonal projections.

Claim. If R is large enough,

$$(N_{n,R}, L_{n,R}) := ((\overline{B_R} \cap E_n^+) \times (\overline{B_R} \cap E_n^-), (\overline{B_R} \cap E_n^+) \times (\partial\overline{B_R} \cap E_n^-))$$

is an index pair for $I(\overline{N_{n,R} \setminus L_{n,R}})$ w.r.t. $-\nabla_E f_n$ and all critical points of f_n are contained in $N_{n,R}$.

Clearly, $f(N_{n,R} \setminus L_{n,R})$ is bounded.

By (11) there is constant $c > 0$ s.t.

$$\|\nabla_E b(x)\|_E \leq \frac{1}{4\sqrt{2}\|S_0^{-1}\|_{\mathcal{L}(E)}} \|x\|_E + c.$$

If $x \in L_{n,R}$, by $(**)$ we get

$$\begin{aligned}\langle \nabla_E f_n(x), P_n^- x\rangle_E &= \langle P_n S_0 x + P_n \nabla_E b(x), P_n^- x\rangle_E \\ &\leq -\frac{\|P_n^- x\|_E^2}{2\|S_0^{-1}\|_{\mathcal{L}(E)}} + \frac{\|x\|_E \|P_n^- x\|_E}{4\sqrt{2}\|S_0^{-1}\|_{\mathcal{L}(E)}} + c\|P_n^- x\|_E \\ &\leq -\frac{R^2}{2\|S_0^{-1}\|_{\mathcal{L}(E)}} + cR\end{aligned}$$

which is negative for R large enough. Therefore $\Phi_t^{-\nabla_E f_n}(x) \notin N_{n,R}$ for $t > 0$ small enough if $x \in L_{n,R}$, and this implies that $L_{n,R}$ is positively invariant in $N_{n,R}$.

Analogously, if $x \in \partial N_{n,R} \setminus L_{n,R}$,

$$\langle \nabla_E f_n(x), P_n^+ x\rangle_E > 0,$$

so $\Phi_{-t}^{-\nabla_E f_n}(x) \notin N_{n,R}$ for $t > 0$ small enough. This together with the previous fact, $\Phi_t^{-\nabla_E f_n}(x) \notin N_{n,R}$ for $t > 0$ small enough if $x \in L_{n,R}$, implies that $L_{n,R}$ is an exit set for $N_{n,R}$.

This verifies the claim.

Since $(N_{n,R}, L_{n,R})$ is continuously retractable onto $(\overline{B_R} \cap E_n^-, \partial\overline{B_R} \cap E_n^-)$, then

$$H_k(N_{n,R}, L_{n,R}) \cong H_k(\overline{B}^{\dim E_n^-}, \partial\overline{B}^{\dim E_n^-}) = \begin{cases} \mathbb{Z} & k = \dim E_n^- = i(P_n S_0 P_n) \\ 0 & \text{otherwise} \end{cases}$$

Therefore, by Theorem 3.3.3, the Morse relations for f_n are

$$\sum_{x \in \mathrm{Crit}(f_n)} t^{m(x;f_n)} = t^{i(P_n S_0 P_n)} + (1+t)Q_n(t).$$

where $Q_n(t)$ is a polynomial with $\mathbb{Z}_{\geq 0}$-coefficients.

Then $(*)$ implies the Morse relations for f:

$$\sum_{x \in \mathrm{Crit}(f)} t^{m_{E^-(S_0)}(x;f)} = 1 + (1+t)Q(t)$$

with $Q(t) := t^{-i(P_n S_0 P_n)} Q_n(t)$. □

4 Lusternik–Schnirelmann theory

4.1 Lusternik–Schnirelmann category

Definition 4.1.1. Let X be a topological space. The **Lusternik-Schnirelmann category** of a closed subset A of X is defined by

$$\operatorname{cat}_X(A) = \min\{k \,|\, A \text{ can be covered by } k \text{ contractible closed subsets in } X\}$$

We write $\operatorname{cat}(X) := \operatorname{cat}_X(X)$.

Proposition 4.1.2. *The following properties hold:*

(a) $\operatorname{cat}_X(A) = 0 \Leftrightarrow A = \varnothing$.

(b) $\operatorname{cat}_X(A) = 1 \Leftrightarrow \overline{A}$ *is contractible in* X.

(c) $\operatorname{cat}_X(A) = \operatorname{cat}_X(\overline{A})$.

(d) **(monotonicity)** $A \subseteq B \subseteq X \Rightarrow \operatorname{cat}_X(A) \leq \operatorname{cat}_X(B)$.

(e) **(subadditivity)** $\operatorname{cat}_X(A \cup B) \leq \operatorname{cat}_X(A) + \operatorname{cat}_X(B)$.

(f) If A *is closed and* $A \hookrightarrow X$ *is homotopic to a map* $A \to B$, *then* $\operatorname{cat}_X(A) \leq \operatorname{cat}_X(B)$.

(g) $\operatorname{cat}_X(h(A)) = \operatorname{cat}_X(A)$ *for every homeomorphism* $h : X \to X$.

Proposition 4.1.3. *Let* $\mathfrak{M}$ *be a Hilbert manifold and* $A \subseteq \mathfrak{M}$ *a compact subset. Then*

$$\operatorname{cat}_{\mathfrak{M}}(A) \leq \dim A + 1.$$

Theorem 4.1.4 (Lusternik-Schnirelman [Cha05, Theorem 5.2.7])**.** *Let* $\mathfrak{M}$ *be a smooth Banach-Finsler manifold. Suppose* $f \in C^1(\mathfrak{M}, \overline{\mathbb{R}})$ *is bounded from below satisfying* **(PS)**. *Then*

$$\operatorname{Crit}(f) \geq \operatorname{cat}(\mathfrak{M}).$$

4.2 Cup-length estimate

Definition 4.2.1. The cup-length of topological space X is defined as

$$cl(X) = \max\{k \,|\, \exists a_1, \ldots, a_k \in H^*(X) \text{ with } \deg(a_i) \geq 1 \text{ s.t. } a_1 \smile \cdots \smile a_k \neq 0\}$$

For manifold M, then $cl(M) \leq \dim M + 1$.

Lemma 4.2.2. let M and N be manifolds,

$$cl(M \times N) \geq cl(M) + cl(N).$$

Proposition 4.2.3 ([Cha93, Corollary 3.6])**.**

$$\operatorname{cat}(X) \geq cl(X) + 1.$$

Corollary 4.2.4 ([Cha93, Corollary 3.4])**.** *Suppose $f \in C^1(\mathfrak{M}, \overline{\mathbb{R}})$ is bounded from below satisfying* **(PS)**. *Then*

$$\operatorname{Crit}(f) \geq cl(\mathfrak{M}) + 1.$$

As a consequence:

Theorem 4.2.5 ([Abb01, Theorem 4.1.5])**.** *Let $f \in C^2(\mathfrak{M})$ and suppose (N, L), where $f(N \setminus L)$ is bounded, is an index pair for the isolated invariant set $S := I(\overline{N \setminus L})$ w.r.t. $\Phi_t^{-\nabla f}$. Assume that f satisfies* **(PS)** *on $\overline{f(N \setminus L)}$. Then*

$$\operatorname{Crit}(f|_{N \setminus L}) \geq cl(N \setminus L) + 1.$$

An application observed by Arnold:

Theorem 4.2.6 ([HZ94, §6.1, Proposition 1])**.** *Let $\varphi \in \operatorname{Diff}_0(T^2, dx \wedge dy)$ be C^1-close to identity, and satisfying*

$$\int_{T^2} (\varphi(x, y) - (x, y)) dx \wedge dy = (0, 0).$$

Then

$$\#\operatorname{Fix}(\varphi) \geq 3.$$

If φ is furtehr non-degenerate, then

$$\#\operatorname{Fix}(\varphi) \geq 4.$$

Proof. Since

$$\varphi : (x, y) \mapsto (X, Y)$$

is C^1-close to id, it admits generating function $G : \mathbb{R}^2 \to \mathbb{R}$ in the following sense:

$$X = x + \frac{\partial G}{\partial Y}(x, Y)$$
$$Y = y + \frac{\partial G}{\partial x}(x, Y)$$

This yields that ∇G is periodic. It follows from the center of mass preserving condition in the statement that G is periodic and hence descends to a function on T^2. In conclusion, there is one-to-one correspondence:

$$\mathrm{Fix}(\varphi) \overset{1;1}{\longleftrightarrow} \mathrm{Crit}(G)$$

By Lusternik-Schnirelman theorem,

$$\mathrm{Crit}(G) \geq cl(T^2) + 1 = 2 + 1 = 3.$$

If φ is non-degenerate, then F is Morse, and by Morse inequality,

$$\mathrm{Crit}(G) \geq \sum_{k=0}^{2} b_k(T^2) = 1 + 2 + 1 = 4.$$

This proves the statement. □

5 Arnold conjecture for T^{2n}

5.1 Non-degenerate Arnold conjecture

Theorem 5.1.1 ([CZ83]). *Let $\varphi_H \in \mathrm{Ham}(T^{2n}, \omega_0)$ be non-degenerate. Then*

$$\#\operatorname{Fix}(\varphi_H) \geq \sum_{k=1}^{2n} b_k(T^{2n}) = 2^{2n}.$$

Theorem 5.1.2 ([Abb01, Theorem 6.1.6]). *Suppose all $\operatorname{Crit}(\mathcal{A}_H)$ are non-degenerate, then*

$$\sum_{x \in \operatorname{Crit}(\mathcal{A}_H)} t^{m_{\mathbb{R}^n \times E^-}(x; \mathcal{A}_H)} = \sum_{k=0}^{2n} \binom{2n}{k} t^{k-n} + (1+t)Q(t)$$

for some Q is a Laurent polynomial with $\mathbb{Z}_{\geq 0}$-coefficients.

Proof. Since $\nabla_{H^{\frac{1}{2}}} b$ is compact and bounded, $\mathcal{A}_H$ satisfies **(PS)*** on $\overline{\mathbb{R}}$: every sequence $(x_m) \subset \Lambda$ such that $x_m \in \Lambda_m$ and $P_m \nabla_{H^{\frac{1}{2}}} \mathcal{A}_H(x_m) \xrightarrow{m \to \infty} 0$ is sequential compact. In particular, since we assume all critical points of $\mathcal{A}_H$ are non-degenerate, $\operatorname{Crit}(\mathcal{A}_H)$ is finite set. Then Lemma 2.2.10 implies that if n is large enough all critical points of $\mathcal{A}_H|_{\Lambda_n}$ are non-degenerate and there is a bijection

$$j_n : \operatorname{Crit}(\mathcal{A}_H) \to \operatorname{Crit}(\mathcal{A}_H|_{\Lambda_n})$$

s.t.

$$m_{\mathbb{R}^n \times E^-}(x; \mathcal{A}_H) = m(j_m(x); \mathcal{A}_H|_{\Lambda_m}) - \dim F_m^- - n.$$

If m and R are large enough, we find an index pair for $\mathcal{A}_H|_{\Lambda_m}$ as in the proof of 3.3.3:

$$N_{m,R} := T^{2n} \times ((\overline{B}_R \cap F_m^+) \times (\overline{B}_R) \cap F_m^-)$$
$$L_{m,R} := T^{2n} \times ((\overline{B}_R \cap F_m^+) \times (\partial \overline{B}_R) \cap F_m^-)$$

Then by Theorem 3.3.3 and Künneth formula, the Poincaré polynomial of $(N_{m,R}, L_{m,R})$ is

$$P(t, N_{m,R}, L_{m,R}) = t^{\dim F_m^-} P(t, T^{2n}) = \sum_{k=0}^{2n} \binom{2n}{k} t^{k + \dim F_m^-}.$$

Therefore, the Morse relation for $\mathcal{A}_H|_{\Lambda_n}$ is

$$\sum_{x \in \operatorname{Crit}(\mathcal{A}_H|_{\Lambda_m})} t^{m(x; \mathcal{A}_H|_{\Lambda_m})} = \sum_{k=0}^{2n} \binom{2n}{k} t^{k + \dim F_m^-} + (1+t)Q_m(t).$$

The Morse relation for $\operatorname{Crit}(\mathcal{A}_H)$ follows with $Q(t) := t^{-\dim F_m^- - n} Q_m(t)$. □

The Theorem 5.1.1 is then a direct consequence.

5.2 Degenerate Arnold conjecture

Theorem 5.2.1 ([CZ83]). *Let $\varphi_H \in \mathrm{Ham}(T^{2n}, \omega_0)$ be general. Then*

$$\#\,\mathrm{Fix}(\varphi_H) \geq 2n + 1.$$

The Galerkin reduction used in last section might not be sufficient to deal with a functional which may have degenerate critical points: even if one shows the existence of $2n+1$ critical points for $\mathcal{A}_H|_{\Lambda_m}$, one would have to guarantee that taking limit $m \to \infty$ there are still $2n+1$ distinct critical points of $\mathcal{A}_H$. Therefore, we apply a slightly different approach known as saddle point reduction.

Proof. The set

$$\left\{\nabla^2_{H^{\frac{1}{2}}} b(x) : x \in \Lambda\right\}$$

is a relatively compact subset of the space of compact operators on E. This implies that we can find a positive integer m s.t. for every $x \in \Lambda$ the following estimate holds:

$$\left\|(I - P_m)\nabla^2_{H^{\frac{1}{2}}} b(x)\right\|_{\mathcal{L}(E)} < \frac{1}{2} \tag{14}$$

We look for the critical points of $\mathcal{A}_H$ in the subset $\Psi^{-1}(0)$ where

$$\begin{aligned} \Psi : \Lambda &\to F_m^\perp \\ x &\mapsto (I - P_m)\nabla_{H^{\frac{1}{2}}} \mathcal{A}_H(x) \end{aligned}$$

The above estimate guarantees that $\Psi^{-1}(0)$ is $\mathrm{graph}(\Phi)$ of $\Phi : \Lambda_m \to F_m^\perp$. Indeed, given $x \in \Lambda_m$, every $y \in F_m^\perp$ s.t. $(I - P_m)\nabla_{H^{\frac{1}{2}}} \mathcal{A}_H(x) = 0$ is a fixed point of the map

$$\begin{aligned} \psi_x : F_m^\perp &\to F_m^\perp \\ y &\mapsto -S^{-1}(I - P_m)\nabla_{H^{\frac{1}{2}}} b(x + y) \end{aligned}$$

where $S := P^+ - P^-$ is the invertible self-adjoint oerpator in the first term of the action functional (8). Since $\|S^{-1}\|_{\mathcal{L}(E)} = 1$, the estimate (14) implies that ψ_x is contraction, hence it has a unique fixed point by Banach fixed point theorem, and $\Psi^{-1}(0) = \mathrm{graph}(\Phi)$ for some $\Phi : \Lambda_m \to F_m^\perp$. Moreover, the differential of Ψ given by

$$d\Psi(x) = S(I - P_m) + (I - P_m)\nabla^2_{H^{\frac{1}{2}}} b(x)$$

maps $F_m^\perp$ onto $F_m^\perp$. By implicit function theorem, Φ is smooth. Furthermore, $\|d\Phi\|_{\mathcal{L}(E)}$ is uniformly bounded for $x \in \Lambda_m$.

Let $f(x) \in C^\infty(\Lambda_m)$ be the function

$$f(x) := \mathcal{A}_H(x + \Phi(x)) \qquad \forall x \in \Lambda_m.$$

The map $\mathrm{id} + \Phi$ provides with a bijection

$$\mathrm{Crit}(f) \cong \mathrm{Crit}(\mathcal{A}_H).$$

Therefore it is enough to prove that

$$\#\,\mathrm{Crit}(f) \geq 2n + 1.$$

The functional $\mathcal{A}_H$ satisfies **(PS)** on $\overline{\mathbb{R}}$ since S is invertible and $\nabla_{H^{\frac{1}{2}}} b$ is compact and bounded. The indentity

$$\nabla_{H^{\frac{1}{2}}} f(x) = (I + d\Phi(x))^* \nabla_{H^{\frac{1}{2}}} \mathcal{A}_H(x + \Phi(x))$$

together with the facts that $\nabla_{H^{\frac{1}{2}}} \mathcal{A}_H(x + \Phi(x)) \in \mathbb{R}^{2n} \oplus F_m$ and

$$(I + d\Phi(x))^* : \mathbb{R}^{2n} \oplus F_m \to T_x \Lambda_m = E_m$$

has bounded inverse, uniformly w.r.t. $x \in \Lambda_m$, easily imply that f also satisfies **(PS)** on $\overline{\mathbb{R}}$. Similarly, the fact that $\nabla_{H^{\frac{1}{2}}} b$ is bounded implies that

$$N_{m,R} := T^{2n} \times ((\overline{B}_R \cap F_m^+) \times (\overline{B}_R) \cap F_m^-)$$
$$L_{m,R} := T^{2n} \times ((\overline{B}_R \cap F_m^+) \times (\partial \overline{B}_R) \cap F_m^-)$$

is an index pair for f, provided R is large enough. We have cup-length

$$cl(N_{m,R} \setminus L_{m,R}) = cl(T^{2n}) = 2n.$$

By Theorem 4.2.5, $\#\,\mathrm{Crit}(f) \geq 2n + 1$. □

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